

Read-Heavy System Capacity Example

Read-Heavy System Example (URL Shortener)

Assumptions:

- DAU = 10M
- Redirects per user per day = 100
- Shortens per user per day = 0.01 (100K/day total)
- Request size = 500 bytes (lookup), 200 bytes (log)

Step 1. RPS:

- Redirects: $(10M \times 100) / 86,400 \approx 11.5K$ RPS
- Shortens: $100K / 86,400 \approx 1.15$ RPS

Step 2. Throughput:

- Redirects: $11.5K \times 500B \approx 5.75$ MB/sec
- Logs: $1.15K \times 200B \approx 0.23$ MB/sec

Step 3. Daily Storage:

- Logs: $0.23 \text{ MB/sec} \times 86,400 \approx 19.9$ GB/day

Step 4. Peak Load:

- Redirects $\approx 34.5K$ RPS (3× avg)

Key Observation:

- Reads dominate massively, writes negligible.